

E12 ADDITIONAL DATA Multiple Reactions in a Nonisothermal PFR

Why Are We Learning This?

So far, we have solved energy balances for single reactions. In practice, however, many reactors contain **multiple reactions occurring simultaneously**, and each reaction may have a different effect on reactor temperature. Some reactions release heat, while others absorb heat. Understanding how to account for these competing effects is essential for predicting reactor performance and maintaining safe operation.

In this lesson, we learn how:

- Multiple reactions contribute to overall species generation and consumption.
- Relative rates can be used to convert between different species bases.
- Heat effects from multiple reactions are combined in an energy balance.
- Sign conventions influence the interpretation of exothermic and endothermic reactions.
- Physical intuition can be used to check whether an energy balance is written correctly.
- Multiple reactions can simultaneously heat and cool a reactor.

By understanding these ideas, we can develop energy balances that accurately predict reactor temperature and avoid common sign-convention mistakes.

Course Roadmap

Introduction to CRE



General Balance Equation



Common Reactor Models
(CSTR, Batch, PFR)



Reaction Progress

(Conversion)



Graphical Reactor Design
(Inverse Rate/Levenspiel Plots)



Rate Laws
(Kinetics)



Equilibrium
(Reversible Reactions)



Equilibrium Conversion
(Thermodynamic Limitations)



One Reaction: Solving in terms of Reaction Progress



One Reaction: Reactor Design
(CSTR, Batch, PFR, PBR)



Catalytic Reactions



Numerical Methods
(Python)



Energy Balances



One Reaction: Non-Isothermal Reactor Design



► Multiple Reactions ◀



Reactor Safety



Challenging the Well-Mixed Assumption
(Residence Time Distributions)



Challenging the Elementary Rate Law Assumption
(Reaction Mechanisms)

Learning Objectives

After completing this lesson, you should be able to:

- ✓ Write mole balances for reactors involving multiple reactions.
- ✓ Determine overall species generation rates by combining contributions from multiple reactions.
- ✓ Use stoichiometry and relative rates to convert between different species rate expressions.
- ✓ Write energy balances for systems containing multiple reactions.
- ✓ Correctly apply heats of reaction for both exothermic and endothermic reactions.
- ✓ Explain why reactant-based and product-based formulations produce the same physical result when applied consistently.
- ✓ Predict whether individual reactions will increase or decrease reactor temperature.
- ✓ Use physical reasoning and temperature trends to identify sign errors in energy balances.
- ✓ Develop coupled mole and energy balances for non-isothermal reactors with multiple reactions.

Big Idea

When multiple reactions occur simultaneously:

Reaction 1
↓
Heat Effect 1

Reaction 2

↓
Heat Effect 2

Reaction 3
↓
Heat Effect 3

All reaction contributions must be combined consistently in the energy balance.

Reaction Rates
+
Heats of Reaction
↓
Energy Balance
↓
Function That Describes Temperature Change

A well-written energy balance should always predict the correct physical behavior:

Exothermic Reaction
↓
Temperature Increases

Endothermic Reaction
↓
Temperature Decreases

Multiple Reactions Require Multiple Rate Contributions

When a species participates in more than one reaction, its overall rate of generation must include contributions from all reactions involving that species.

For example:

Species A

↓
 Reaction 1 Contribution
 +
 Reaction 2 Contribution
 ↓
 Overall Rate of A

A species that appears in multiple reactions will generally have multiple terms in its mole balance.

Writing Mole Balances for Multiple Reactions

For a PFR with multiple reactions:

$$\frac{dF_i}{dV} = r_{net,i}$$

where:

$$r_{net,i} = \sum_j \nu_{ij} r_{ij}$$

Where j is over reactions.

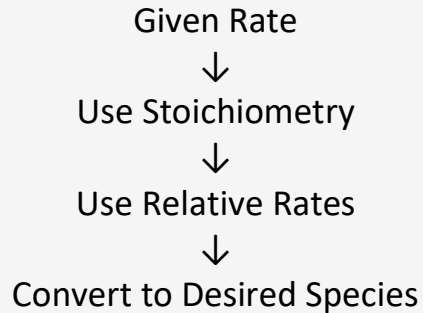
The overall rate of generation of a species is found by summing contributions from every reaction in which that species participates.

Species A
 ↓
 Reaction 1
 +
 Reaction 2
 ↓
 $r_{net,A}$

Relative Rates Are an Accounting Tool

Rate laws are not always given in terms of the species needed for the balance.

In that situation:



Relative rates allow us to move between species while preserving stoichiometric consistency.

Consistency Matters More Than Form

There are several valid ways to write mole and energy balances.

The important requirement is: All Terms Use A Consistent Basis and Correct Physical Behavior

Different formulations should produce:

- The same numerical solution
- The same reactor behavior
- The same temperature trends

Choosing a Basis for the Energy Balance

Many students find it easiest to write energy balances using rates based on reactant species.

If reactant species are used: Rate Term Sign and Heat of Reaction Sign must be handled consistently.

A consistent sign convention ensures that:

- Exothermic reactions increase temperature.
- Endothermic reactions decrease temperature.

Sanity Check Your Energy Balance

Before trusting an equation, ask: What Should Happen Physically?

For example:

Exothermic Reaction
↓
Temperature Should Increase

and

Endothermic Reaction
↓
Temperature Should Decrease

If your equation predicts the opposite behavior, a sign error may be present.

Exothermic Reactions Add Energy

For an exothermic reaction:

$$\Delta H_{rxn} < 0$$

As reaction proceeds:

Heat Released
↓
Energy Added To Reactor
↓
Temperature Increases

A correctly written energy balance should reproduce this behavior automatically.

Endothermic Reactions Remove Energy

For an endothermic reaction:

$$\Delta H_{rxn} > 0$$

As reaction proceeds:

Energy Consumed



Energy Removed



Temperature Decreases

A correctly written energy balance should predict reactor cooling as the reaction progresses.

Reactants and Products Behave Differently

The sign of a species generation term depends on whether the species is a:

Reactant

or

Product

For a forward reaction:

Reactants



Consumed



Negative Rate of Generation

Products



Formed



Positive Rate of Generation

This difference must be reflected in the energy balance.

Physical Interpretation Is a Powerful Error Check

After writing an energy balance, ask:

Reaction 1



Should Reactor Get Hotter?

Reaction 2



Should Reactor Get Cooler?

If the mathematical model predicts the expected temperature trends, the signs are likely correct.

If not: Check Basis, Check Signs, Check Relative Rates

Key Takeaway

For multiple-reaction energy balances:

Reaction Stoichiometry



Relative Rates



Species Generation Terms



Energy Balance



Temperature Change

It is important to check that the energy balance predicts the correct physical effect of each reaction on reactor temperature.